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BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *Predictive Urban Drainage Blockage and Flash Flood Warning Node*

by

Group_No: 02

Section: 08

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Abstract

Water levels should be controlled very well, particularly in flood prone regions such as Bangladesh. In this project we constructed an Arduino Uno based smart water pumping system which is able to detect the level of water automatically and make actions to avoid water and flood damages. The system consists of an ultrasonic sensor to determine the water level, and an IR sensor to know whether someone or an object is close to the system. According to sensor indications, the system illuminates various colored LEDs to indicate the status. At high water level, a servo motor opens a bypassing path allowing some water to pass away. A vibration motor is triggered when the water level is dangerous to cause the water to shake off any debris that may be blocking the water, and a water pump attached to a motor driver automatically pumps out the excess water. It transmits wireless signals to a device in the vicinity via a Bluetooth module hence the user is always aware of the system status. This project shows that simple embedded systems can be adopted to automate water management in a cost effective and simple manner.

Keywords

Arduino, ultrasonic sensor, IR sensor, servo motor, Bluetooth, water level monitoring, embedded systems, automation.

1. Introduction

The issue of water management is one of the most significant ones nowadays. When it pours heavily as it has been in countries such as Bangladesh, the chances of floods are high and the lack of proper water drainage systems worsens the situation. Most of the water pump systems that are in use today are mechanical in nature and this implies that someone must be around to monitor the water level and make a decision as to whether to switch on or off. This is infeasible particularly in the case of emergencies.

We hope to address this issue by constructing an automatic smart water pumping system in our project. The system has sensors to monitor the water level and will take the appropriate action without human intervention. It also has an IR sensor to identify the presence of a person or object in the vicinity, which provides an added safety measure. The system gives real time alerts through Bluetooth such that the user is always aware of what is happening. It discusses such things as sensor interfacing, actuator control, motor driving, and wireless communication.

The principal aims of this project are:

- To automatically measure the water level with the help of an ultrasonic sensor.
- To indicate the users with LEDs depending on the water level condition.
- To drive a servo motor, vibration motor and water pump using sensor data.
- To identify nearby objects with the help of an IR sensor and notify users.
- To transmit wireless status alerts with a Bluetooth module.

2. Related Work/ Inspiration

Some water level monitoring systems are already in place though primarily in industrial environments. A lot of these systems have float sensors or pressure sensors to gauge water levels. These systems are however too costly or too complicated to be used on a small scale. Arduino combined with ultrasonic sensors has been used in some research projects to sense water levels, although most of these projects simply show the water level on an LCD display, without necessarily performing any automated action.

Our other inspiration was provided by flood early warning systems in certain areas of Bangladesh, whereby sensors are fitted on rivers in order to provide warning when the water levels become excessively high. Our project uses a similar concept but on a smaller scale such as a water tank or drainage system.

The uniqueness of our project is that it is not only a monitor of the water level, but also makes actions on it. The servo motor provides an alternate route, the vibration motor sweeps away debris and the pump drains off the surplus water. All these automatically occur without the user having to take any action. The Bluetooth alert system is an additional useful feature since the user does not need to be at the physical location to be aware of what is going on.

3. Technical Approach

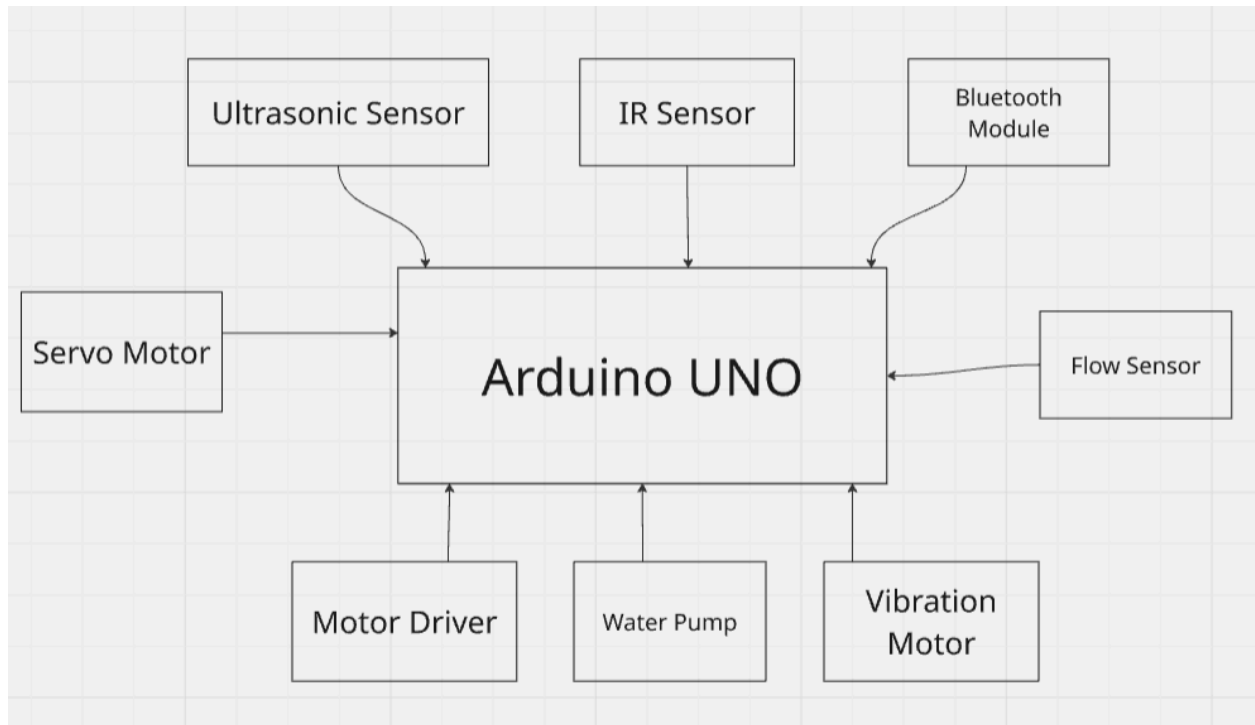
System Architecture:

It is constructed using an Arduino Uno, the brain of the whole project. The Arduino is linked to all the sensors and actuators. The sensors are read by the Arduino and the action is taken depending on the water level and the sensing IR results. The structure of the system is described below:

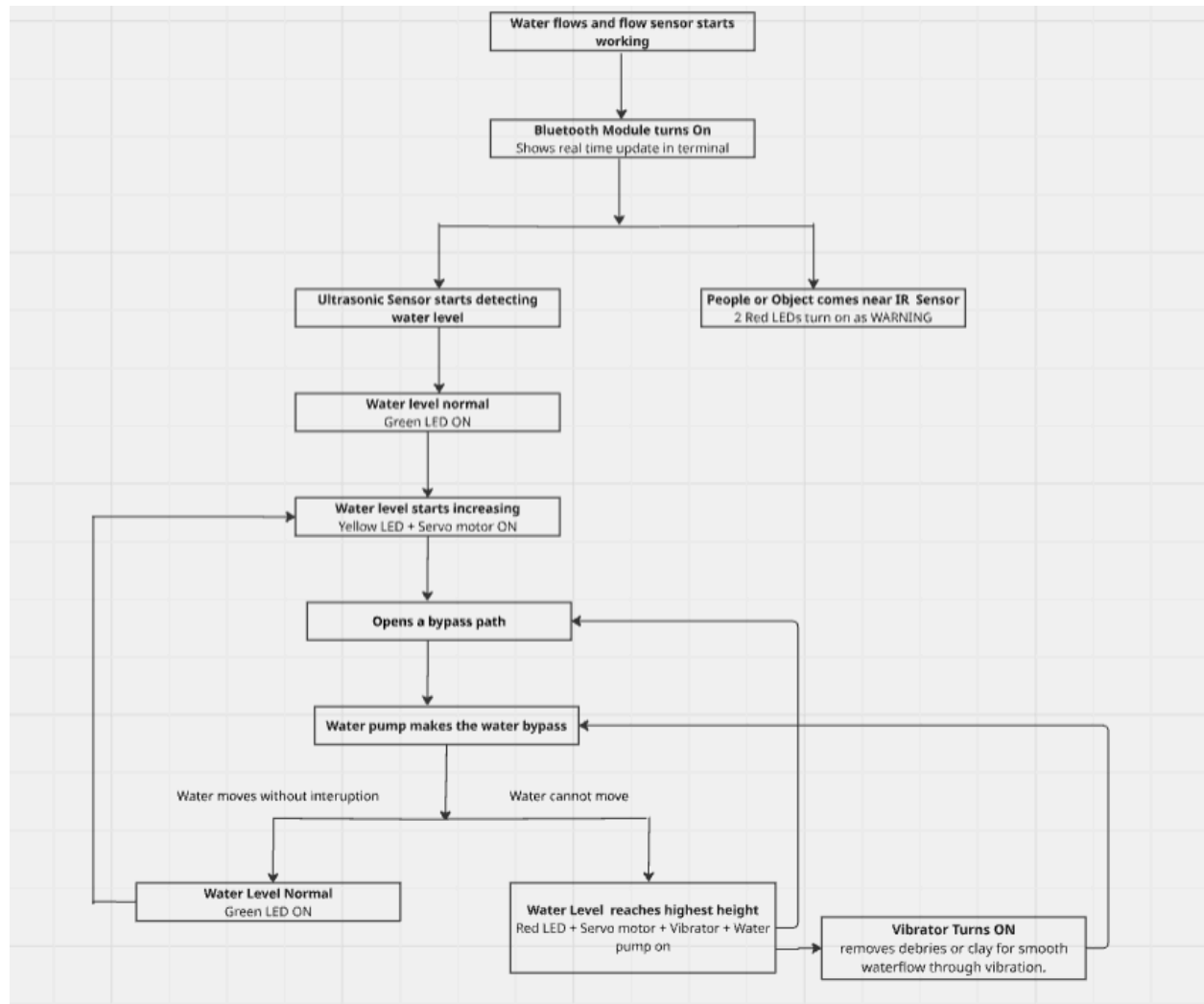
- The sensor of the IR is kept close to the system in order to detect any human or object. Two red LEDs are lit when it detects something as a warning.
- The ultrasonic sensor is used to measure the distance between itself and the water surface. The lesser the distance, the greater the water level.
- The Arduino makes the relevant LED (green, yellow or red) light up and switches on the relevant component based on the level of the water.
- The Bluetooth module (HC-05) transmits status messages to a connected smartphone or device.
- The flow sensor measures the flow of water.

The general processing is, data is read by Sensor and sent to Arduino, which then processes it and sends it to LED which lights up and then to the Bluetooth module, which then sends a Bluetooth alert.

Block Diagram:



Flowchart:



Components Used:

- Arduino Uno: The main microcontroller of the system.
- HC-SR04 Ultrasonic Sensor: It measures the water level of the water surface by calculating the distance.
- Red LEDs: One LED lights up when the water level reaches a dangerous point and 2 LEDs light up when an object or people come close to the IR sensor.
- Green LED: It lights up when the water is at normal level.
- Yellow LED: It lights up when the water level is higher than normal.
- SG90 Servo Motor: When the yellow light is on, the motor opens a bypass path to make the water flow away.
- YF-S201 Water Flow Sensor: Measures the flow of water.

- Vibration Motor: When the water level is dangerous, and if any debris is blocking the water flow, it starts vibrating.
- IR Sensor Module: When something is detected it turns on the two red LEDs.
- Mini Water Pump: It lets the excess water flow when the water level is dangerous.
- L298N Motor Driver Module: It controls the water pump by providing current from the battery.
- HC-05 Bluetooth Module: It is used to establish serial communication.
- Battery: It provides power to the motor driver for making the water pump work.
- Resistors: It is used to protect the LEDs so that they do not burn out.
- Jumper wires and Breadboard: For building the connections.

Cost Breakdown:

Component	Quantity	Estimated Price (BDT)
Arduino Uno	1	950
HC-SR04 Ultrasonic Sensor	1	110
IR Sensor Module	1	45
YF-S201 Water Flow Sensor	1	300
HC-05 Bluetooth Module	1	380
Vibration Motor	1	150

SG90 Servo Motor	1	180
LEDs (Red, Yellow, Green)	5	80
L298N Motor Driver Module	1	220
Resistor	3	45
Water Pump	1	250
Hose Pipe	1	100
Breadboard	1	100
5V Adapter	1	300
Jumper Wires		150
Battery	1	200

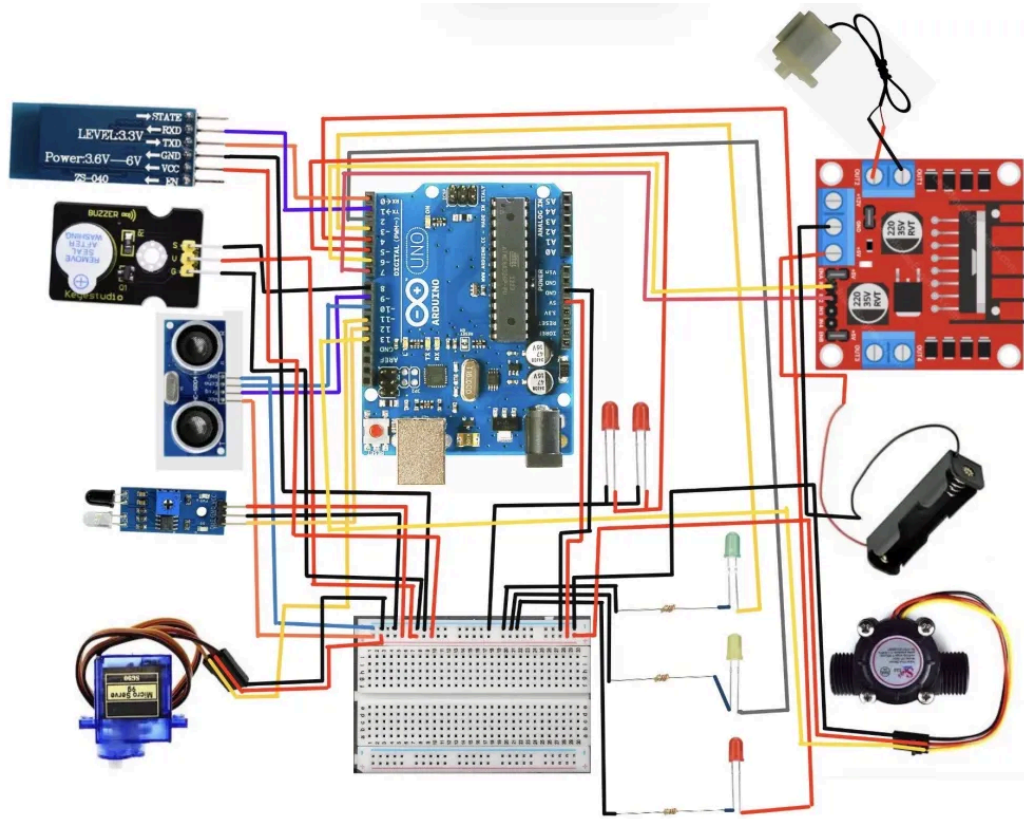
Total Estimated Cost		3560 BDT
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Communication Protocol:

In this project, we employed Bluetooth (serial communication using UART) with the HC-05 module to provide wireless communication. The Arduino is connected to the HC-05 module via SoftwareSerial library at 9600 bps. The module transmits text based status messages to a paired Bluetooth device, for example, a smartphone with a Bluetooth terminal application.

We have selected Bluetooth since it is easy, cheap, and can be used in close range communication. It is also not connected to the internet and this makes it suitable in places where wifi services may not be available. The messages relayed will entail warning messages such as water level normal, water level high, water level critical, and IR object detected.

Circuit/Schematic:



The circuit was developed on a breadboard. The most important links are:

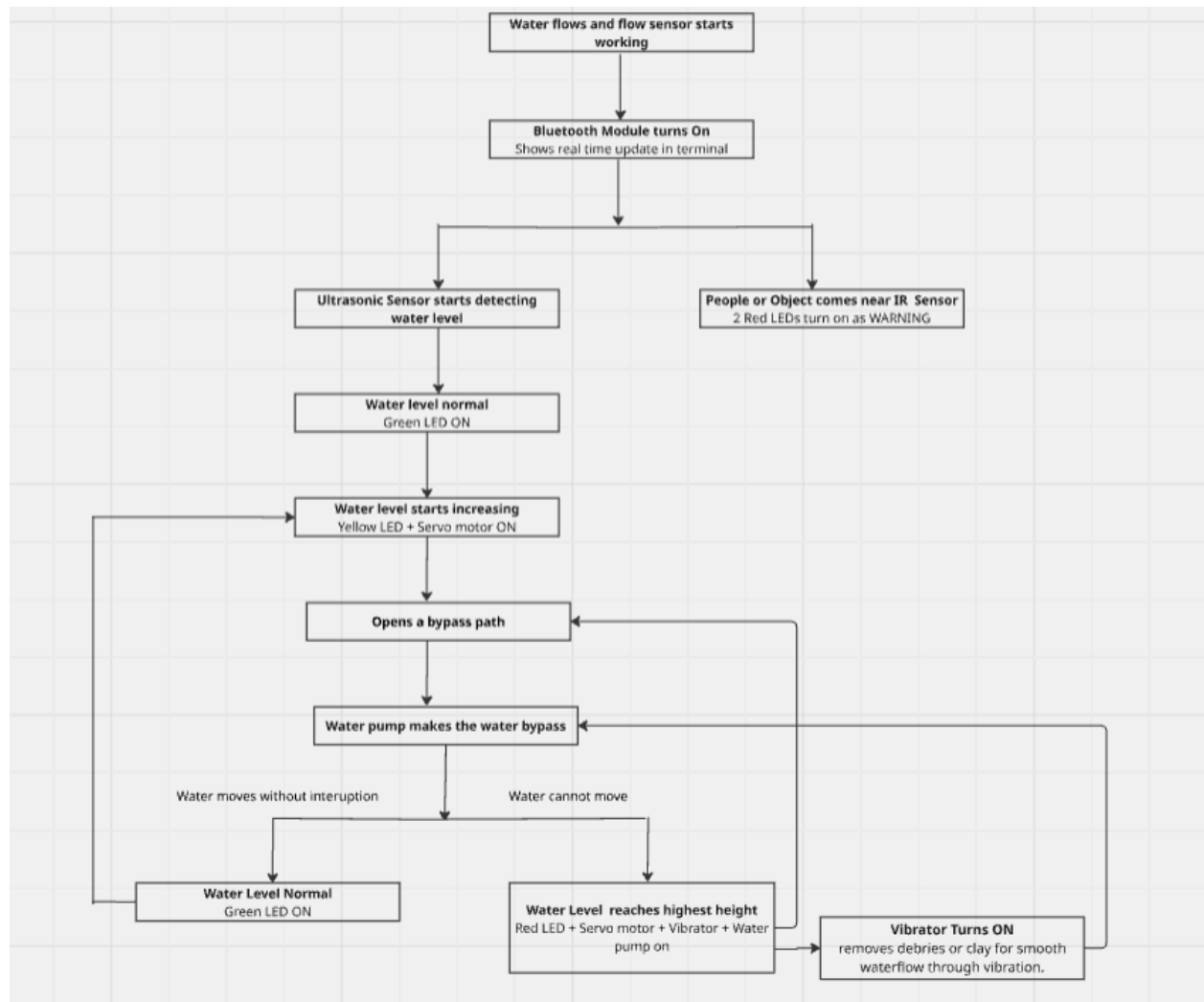
- HC-SR04 ultrasonic sensor is attached to the digital pins 9 (Trigger) and 10 (Echo) of the Arduino.
- IR sensor output pins: The IR sensor output pin is connected to digital pin 12.
- The green LED is attached to pin 2, yellow to pin 3 and the red water level LED to pin 4, all with a resistor of 330 ohms.
- The two IR alert red LEDs are linked to pin 5 each one with a resistor of 330 ohms.
- The servo motor signal wire is attached to pin 11.
- Pin 8 is connected to the vibration motor by a transistor.
- The L298N motor driver is connected to the water pump and is powered by the battery. IN1 and IN2 lead to the motor driver connected to Arduino pin 6 and 7.
- HC-05 Bluetooth module TX pin to the Arduino RX (pin 0) and RX pin to Arduino TX (pin 1) is connected using a voltage divider.
- The water flow sensor is connected to the pin 13.

Implementation:

The following is a step by step procedure of how the system works:

1. Arduino is booted up and everything is set up. The servo motor has been set to default closed and the water flow sensor starts detecting if there is any flow of water.
2. The IR sensor is constantly monitoring the presence of objects or people in the area. When it detects something both red LEDs are on and a Bluetooth notification is given stating Object Detected.
3. The ultrasonic sensor emits a sound pulse and measures the time it takes the sound pulse to reflect off the water surface. This provides us with a distance to the water.
4. When the water level is normal (distance is large) then the green LED is on. A Bluetooth message is sent: Water Level: Normal.
5. When the water level is elevated (distance falls below the normal mark), the yellow LED is on. The servo motor rotates to open the bypass path and to open the water to flow out. A Bluetooth message is sent which is Water Level: High Bypass Opened.
6. In case of the danger of the water level (the distance decreases to a critical point), the red LED is activated. The vibration motor begins vibrating to shake off debris and clear up any blockage. The motor driver also activates the water pump to pump out the unnecessary water. Bluetooth notification DANGER: Water Level Critical is received.
7. This is an endless loop, and the Arduino will monitor sensor readings and update all the parts every half a second or so.

Workflow Diagram:



Challenges:

There were a number of challenges that we encountered during the project:

- Calibration of the ultrasonic sensor: The sensor was initially providing unreliable measurements. We had to change the component and then we were getting correct readings.
- Bluetooth interference: HC-05 module was not showing on the device initially. After changing the wires it worked properly and started showing the outputs in the terminal.
- Servo motor jitter: The servo motor was vibrating slightly when not in operation. but after wiring it again it started working fine.

4. Sustainability & Impact

Sustainability:

This project is a sustainable one. Arduino Uno has very low power requirements, hence can operate on solar power or on a small battery over a long period. The water pump is also activated only when necessary and saves energy as opposed to a pump that operates continuously.

The majority of materials employed in this project can be reused and recycled. The jumper wires and breadboard can be reutilized in different projects. Arduino in itself can be repurposed and programmed. The system also prevents wastage of water as it is automated and minimizes the effects of flooding which has long term environmental impact.

Impact:

This system can positively influence a lot of spheres. This type of low-cost automated pumping system in rural Bangladesh (where drainage systems are frequently substandard and flooding is frequent) could be used to ensure homes, farmland, and stored goods are not damaged by water. The Bluetooth alert system will be used so that even when nobody is monitoring the pump, it will notify the user on his/her phone.

The system can be used to autonomously control irrigation tanks or drainage pits in the industrial or agricultural environment. This saves on human effort and the possibility of human error reduces. Another safety feature incorporated by the IR sensor is the alert when a user gets too close to the water system.

Future Work:

This project can be enhanced in the future in a number of ways:

- Substitute the Bluetooth module with wifi module (such as ESP8266) to have an ability to send alerts via the internet and be able to monitor the system regardless of the location.
- Add a mobile application to display live water level information and a history curve.
- Add a solar panel and transform the system into a complete off grid and self powered system.

Adding the second pump or a larger pump is recommended in the case of systems in which the volume of water is larger.

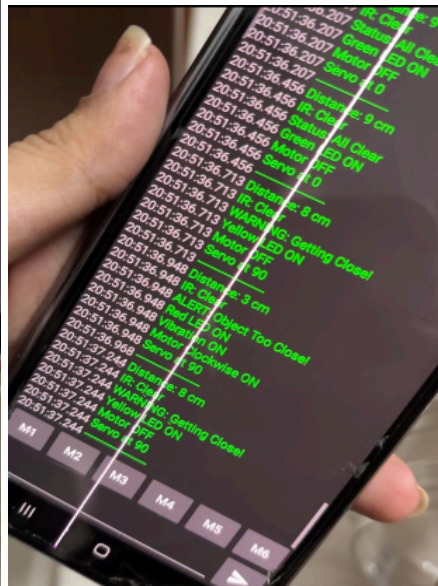
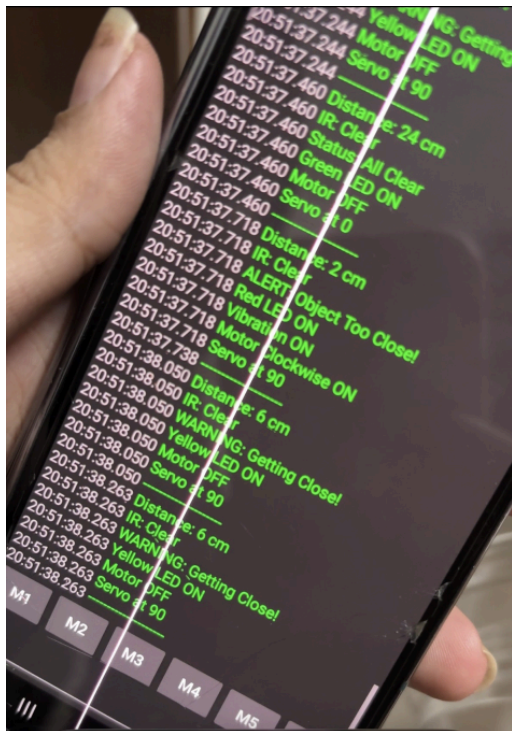
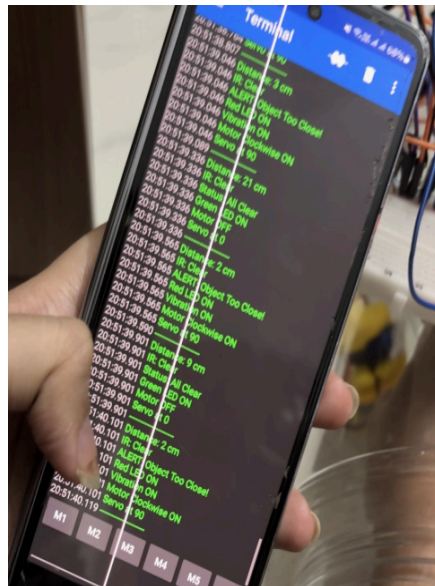
- Add a pH or turbidity sensor to monitor water quality as well as the water level.
- Extend the system to include several water tanks or drain points with one central controller.

Limitations:

The system has certain limitations in spite of being effective in testing:

- The ultrasonic sensor will not indicate the correct values when the water is highly turbulent or in case of foam at the surface.
- Bluetooth has a limited range of approximately 10 meters, and the user has to be close to the device to get a notification.
- The pump that is applied in this project is small and can pump small quantities of water. An actual implementation would require a much more powerful pump.
- The system does not presently have a memory or logging capability. In case of a missed alert, the history can not be viewed.

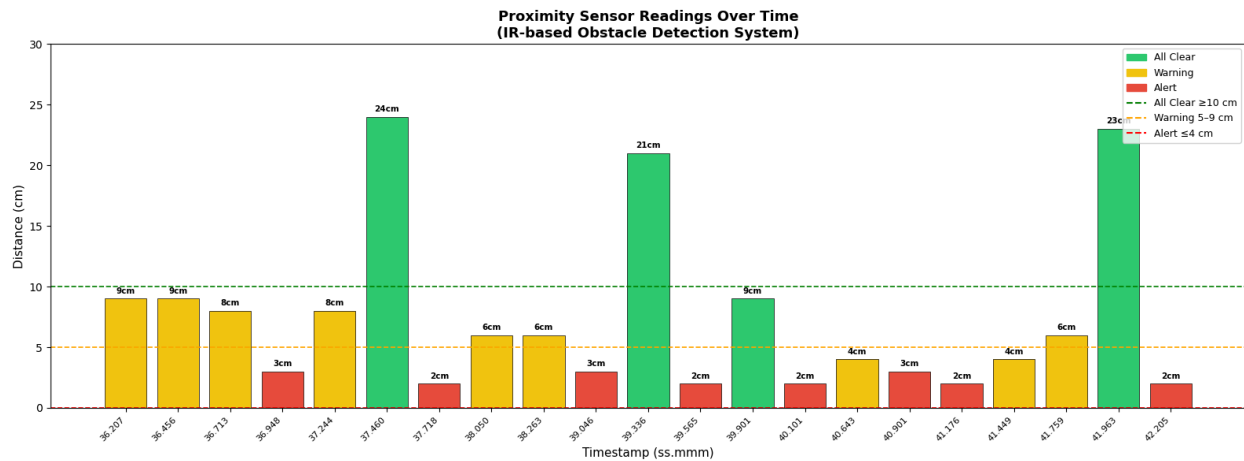
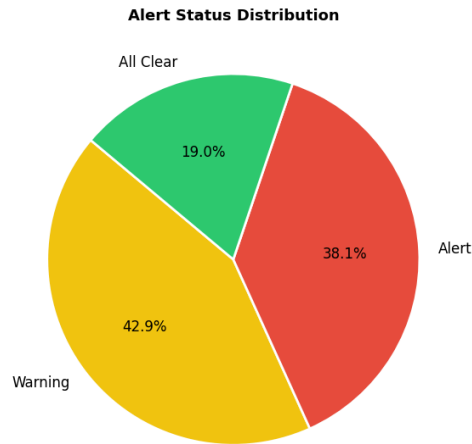
5. Results & Discussion



From the logs, the readings observed were:

Timestamp	Distance	Status
20:51:36.207	9 cm	Warning
20:51:36.456	9 cm	Warning
20:51:36.713	8 cm	Warning
20:51:36.948	3 cm	ALERT
20:51:37.244	8 cm	Warning
20:51:37.460	24 cm	All Clear
20:51:37.718	2 cm	ALERT
20:51:38.050	6 cm	Warning
20:51:38.263	6 cm	Warning
20:51:39.046	3 cm	ALERT
20:51:39.336	21 cm	All Clear

20:51:39.565	2 cm	ALERT
20:51:39.901	9 cm	All Clear
20:51:40.101	2 cm	ALERT
20:51:40.643	4 cm	Warning
20:51:40.901	3 cm	ALERT
20:51:41.176	2 cm	ALERT
20:51:41.449	4 cm	Warning
20:51:41.759	6 cm	Warning
20:51:41.963	23 cm	All Clear
20:51:42.205	2 cm	ALERT



The experiment illustrates an IR-based proximity detector system constructed on the Arduino platform that has been functional for up to about 6 seconds, consisting of 21 sensor measurements. The system was able to categorize the distance of objects into three alert levels namely; a green LED indicating safe distances (>10 cm), a yellow LED with a servo at 90 degree indicating close range warnings (5-9 cm), and a red LED with activation of vibration motor indicating critical proximity warnings (<4 cm). The object remained in the warning and alert areas (81 percent of the readings) most of the time with short gaps of clearance signifying that the sensor was in active operation and was correctly detecting an object that was moving towards it. The system was found to be reliable in its obstacle detecting applications with all hardware responses working as intended, including the LED switching, motor control and servo positioning.

6. Conclusion

This project will demonstrate that we were able to develop a Smart Water Pumping System with an Arduino Uno and a number of sensors and actuators. The system is able to monitor water levels automatically and react to them via a servo motor, vibration motor and a water pump. It also identifies the objects in the vicinity with the help of an IR sensor and transmits real time wireless alerts with the help of a Bluetooth module.

This project demonstrates how simple and inexpensive parts can be integrated to address real life challenges such as floods and inefficient water management. The system is inexpensive, energy efficient and can be expanded to bigger applications. We think that such systems coupled with IoT and cloud connectivity can be used to contribute greatly to the creation of smarter and more resilient water infrastructure in Bangladesh and other places.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Radia Rahman	22299495	Software & Hardware Development	• Arduino programming • IR sensor wiring • Ultrasonic Sensor wiring • LED connection • Motor Driver connection with battery and water pump
Jarin Tasnim Dia	23101261	Software & Hardware Development	• Bluetooth integration • Bluetooth wiring • Vibration Motor wiring • Flow Sensor wiring • Servo motor wiring
Shahadat Hossain	22101559		
Abir Dey	22299046	Project Ideation, Hardware Procurement, and Circuit Assembly	• Conceptualized and proposed the foundational idea for the project. • Sourced and purchased all necessary hardware components for the physical implementation. • Collaborated on the hardware design • Report Writing
Lamia Mahzabin	22201378	Circuit assembly , report writing	• Collaborated on the

Siddiqua			hardware design ● Report Writing
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